Adapting a Topological Framework from Biology to Compositional Crime Dynamics

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I. Motivation

This project is inspired by the paper Topological analysis reveals state transitions in human gut and marine bacterial communities by Chang, VanInsberghe, and Kelly [1]. In that paper, the authors use Mapper to study microbiome dynamics from compositional data. Their framework represents samples as part of a landscape, where dense regions can be interpreted as candidate states and lower-density regions can suggest transitions.

I was interested in this framework because it studies change as movement through a compositional space, rather than only comparing observations through summary statistics or predefined clusters. In the original paper, the main disruptions were biological or environmental, such as cholera recovery, travel-related disturbance, and marine environmental cycles. I wanted to explore whether a similar idea could be adapted to a social phenomenon.

The COVID-19 pandemic was also a period of disruption, but in an urban social system. It affected mobility, reporting patterns, public activity, and the way people used city space. This made me wonder whether the method from Chang et al. [1] could be useful for studying crime composition over time.

The goal is not to claim that crime behaves like a biological system. Instead, the goal is to test whether a framework designed for compositional dynamics can help describe changes in a social dataset. In this project, each community-month is represented by the relative share of different crime types, and the Mapper framework is used to explore whether COVID-19 was associated with changes in the crime-composition landscape.

II. Introduction

During the COVID-19 pandemic, many aspects of urban life changed at the same time. In public discussion, crime during and after the pandemic is often described in terms of whether total crime went up or down. This is useful, but it only captures one part of the story.

A community can have similar total number of reported incidents while still having a different internal crime profile as the example in Figure 1). In that case, total volume alone would not show the full change.

For this reason, this project focuses on crime composition rather than crime volume. I use reported crime incidents from the City of Chicago and aggregate them by community area and month. Each community-month is converted into a compositional vector, where the entries represent the share of each crime type. This makes it possible to compare communities and months based on the structure of reported crime, not only the number of incidents.

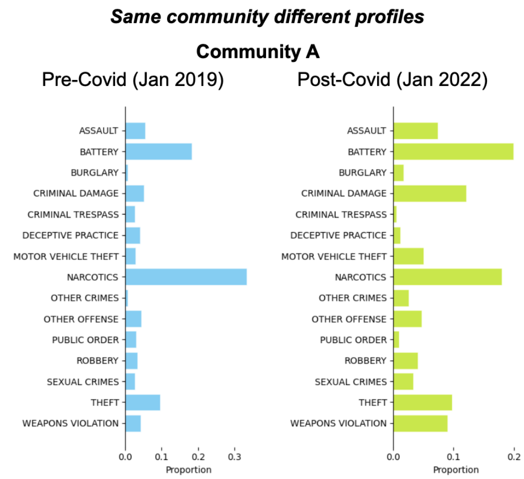

Fig. 1. Composition of crimes from Community A, in different periods. Pre-Covid sample was dominated by Narcotics, also Battery had an important share but visible small than Narcotics. In Post-Covid sample Battery and Narcotics had similar shares, but Narcotics decreased its importance. Additionally, volume was distributed along more categories, than before.

The main research question is:

Did Chicago community areas return to their pre-pandemic crime composition after COVID-19, or did some of them transition into new and persistent compositional regimes?

This question is motivated by the idea that a disruption may not only change the amount of crime, but also the balance between types of crime. A community may appear to "recover" in terms of total volume while still occupying a different region of the crime-composition space. Related questions are whether the COVID period appears as an intermediate region, whether post-COVID observations move back toward pre-COVID patterns, and whether some communities show stronger compositional shifts than others.

III. Background and Related Work

A. Mapper framework to study compositional landscape

This project is based on the framework proposed by Chang et. al [1]. Their paper uses Mapper to study microbiome time series as high-dimensional compositional data. Instead of assigning each sample to a predefined cluster, they represent the data as a graph that preserves local similarity and connectivity across the phase space. In their setting, nodes summarize neighborhoods of similar microbial compositions, and edges connect nodes that share samples. This allows the graph to be interpreted as a landscape, where dense regions can be treated as candidate states and low-density regions can represent transitions. Their framework has two main stages. First, they construct a Mapper graph from compositional samples using a Jensen-Shannon distance matrix, PCoA rank coordinates as the lens, and local clustering inside overlapping regions of the lens. Second, they analyze the graph as a density landscape by estimating node density, identifying local maxima, and assigning states based on graph distance to those maxima. Figure 2 summarizes this workflow from Chang et al. and shows the methodology that I adapted in this project.

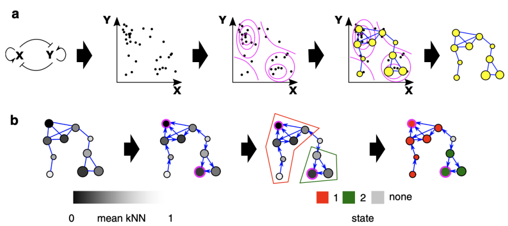

Fig. 2. Methodological framework from Chang et al. [1]. The figure shows the two main stages used in the original paper and adapted here: (a) Mapper graph construction and (b) state assignment based on node density and local maxima.

B. Previous work on COVID-19 and crime

Related to the phenomenon studied here, I found some previous work on crime dynamics during COVID-19. This is not an exhaustive literature review, but it helped me situate the project. For example, Hou et al. [11] studied crime incidents during COVID-19 across several major cities, including Chicago, Los Angeles, New York City, and Washington, D.C. Their results suggest that the pandemic affected crime patterns differently depending on the city and the type of offense. This already points to the idea that COVID-19 did not produce one uniform crime response across urban settings.

I also found work focused specifically on Chicago. Yang et al. [12] studied the impact of COVID-19 on spatial and temporal crime patterns in the city, using approaches such as temporal decomposition, spatial analysis, and comparisons across crime types. Their work also treats the pandemic as a disruption in crime dynamics, but from a more traditional spatial-temporal and statistical perspective.

These papers are useful context because they show that the question of pandemic-related crime change has already been studied, but mostly through changes in counts, trends, spatial concentration, or individual crime categories. My project starts from a related intuition but asks it differently: instead of asking only whether crime increased or decreased, I ask whether the relative composition of crime changed across Chicago community areas. In that sense, I use Mapper to explore whether community-month crime profiles occupy different regions of a compositional landscape before, during, and after COVID-19.

IV. Methods

The following methodology is based on the section ”Methods” in Chang et. al [1] some adaptations were made based on the needs of this project.

Data Preprocessing

Mapper features: The main dataset for this project was the City of Chicago Crimes, 2001 to Present [2] dataset. I used reported incidents from 2018 to 2023 and the variables: date, community area, primary crime type, arrest flag, and domestic flag. I consolidated some of the original crime categories. Less frequent categories, each representing less than 1% of the total records, were grouped into broader labels. This reduced noise from rare labels while keeping the main structure of the crime composition. After pre-processing the crime type label, I aggregated the data by community area and month. For each community-month sample, raw crime counts were transformed into compositional vectors, where each row sums to one. In this representation, each sample describes the internal mix of crime types. I then assigned each sample to one of three periods: Pre-COVID (2018-2019), COVID (2020-2021), and Post-COVID (2022-2023).

Color and descriptive features: The features mentioned in this section were not used to build the Mapper graph, their role was to help describe regions in the Mapper graph.

Calculated arrest rate and domestic violence rate from [2] for each community-month. These variables were not used to build the Mapper graph.

Additionally, Socioeconomic variables were added from the American Community Survey (ACS) [3]. I used community-level ACS features such as poverty rate, unemployment rate, bachelor’s degree or higher rate, renter rate, vacancy rate, median household income, and racial or ethnic population shares. To align ACS information with the crime panel, each community-month was matched to an ACS reference year: 2019 for Pre-COVID observations, 2021 for COVID observations, and 2023 for Post-COVID observations.

Compositional Distance Matrix

Since each community-month vector is converted into relative shares, each row sums to one and can be interpreted as a probability distribution over crime types. Following Chang et al. [1], I used Jensen-Shannon Distance (JSD) (see Appendix A) as the distance between two compositional vectors. This metric was used to build the pairwise distance matrix where each entry compares two community-month samples based on how similar or different their crime composition was.

Mapper

Chang et. al [1] used Mapper to represent the shape of the distribution of data points in high-dimensional phase space as an undirected graph. Where vertices represent neighborhoods of phase space spanned by subsets of adjacent points and the edges represent the connectivity between neighbors.

1) Lense function: PCoA rank coordinates: Following Chang et al. [1], I used Principal Coordinate Analysis (PCoA), also known as classical multidimensional scaling, to project the Jensen-Shannon distance matrix. PCoA is an ordination method developed by Gower [8], [9] that starts from a pairwise distance or similarity matrix and finds coordinates that preserve the main distance relationships among samples. Conceptually, this involves centering the squared distance matrix, applying an eigenvalue decomposition, and using the leading dimensions as coordinates.

I retained the first two PCoA coordinates, rank-transformed them, and used them as the two Mapper filter functions. The rank transformation keeps the ordering of samples along each PCoA axis, while reducing the influence of extreme spacing in the lens values.

For computational efficiency, I used a custom implementation that returns only the first two PCoA coordinates required for the Mapper lens, rather than computing the full coordinate representation.

As in Chang et al., the PCoA coordinates were used only to define the Mapper lens. The local clustering step was still performed using the original Jensen-Shannon distance matrix, because the PCoA projection does not preserve all pairwise distance information.

2) Cover and Local CLustering: After defining the two ranked PCoA coordinates, samples were divided into overlapping intervals along the two lens dimensions. Within each overlapping region, local clustering was performed using single linkage on the corresponding subset of the original distance matrix.

The main adaptation relative to Chang et al. was the cut distance used in the single-linkage step. Chang et al. selected this cutoff locally within each bin based on the distribution of branch lengths. In my data, that local cutoff rule produced many one-sample nodes, which made the graph difficult to interpret. I therefore kept the same single-linkage clustering algorithm, but used a uniform cut distance selected from the overall distance distribution.

The final cutoff parameter was selected based on the rules set for "Hyperparameter selection" describe later in this subsection.

3) Mapper graph: The Mapper graph was constructed using Kepler Mapper in Python. Each node represents a local cluster of community-month samples inside an overlapping region of the lens. An edge connects two nodes when they share at least one sample. For visualization, the graph was converted to a NetworkX object and plotted using `spring_layout`, which implements a Fruchterman-Reingold force-directed layout as in [1]. The visualization algorithm only affects the spatial placement of nodes in the figure, but the analysis relied on the connectivity of the graph, node membership, density, etc.

4) Hyperparameter selection: To select the number of intervals, overlap and single-linkage cutoff, I followed the heuristic rules proposed in [1]:

1) The largest vertex in the final Mapper should represent no more than $\approx 10\%$ of the total samples.
2) The number of connected components representing only one data point should be minimized.

Chang et al. [1] do not provide a fixed numerical threshold for the second rule. For this project, I used 1% of the total samples as upper bound for one-sample components while testing parameter grids. I chose 1% pragmatically: a stricter cutoff would have rejected most parameter combinations, while a looser one would have allowed highly fragmented graphs.

I tested several combinations of intervals, overlap, and single-linkage cutoff distance. The goal was not to optimize the graph to produce a specific result, but to choose parameters that satisfied the methodological rules described above and produced a stable, interpretable Mapper graph. Table I shows the final hyperparameters used for the full dataset and for the exploratory 26-community subset.

TABLE I
FINAL MAPPER HYPERPARAMETERS

Analysis	# intervals	%overlap	cutoff distance
Total dataset	(15,15)	0.3	0.4
26 communities	(10, 10)	0.2	0.45

Density Estimation

After building the Mapper graph, I estimated node density following Chang et al. The goal of this step is to identify regions of the graph where samples are more densely concentrated in the original distance space.

1) Calculate the k-nearest neighbors (kNN) distance for each data point i, and define k-neighborhood $N(k)_i$. The KNN distance of a point i is defined as:
$$\mathrm{kNN}(i,k) = \frac{\sum_{j \in N(k)_i} d_{ij}}{k} \tag{1}$$
where d_{ij} is the distance between points i and j and $N(k)_i$: set of i's nearest neighbors.
Nodes with smaller average kNN distance correspond to denser regions of the sample space.
2) For a vertex V representing n points, the inverse density is defined:
$$D_{\mathrm{inv}}(V) = \frac{\sum_{i \in V} \mathrm{kNN}(i,k)}{n^2} \tag{2}$$
where n^2 compensates the difference of vertices sizes.
3) Finally, the density of a vertex/node is defined as:
$$D(V) = \frac{1}{D_{\mathrm{inv}}}. \tag{3}$$

I applied this method as in Chang et al. with k=10% of the total samples in the dataset.

State Assignment

Chang et al. defined states as topological features of the density surrounding local maxima of density (D). I followed the exact same idea using the following steps:

1) Identify local maxima. Each node whose density was higher than the density of its neighbors was treated as a local maximum. If connected nodes were tied with respect to D, both were assigned as local maximum (potential candidates for state labels).
2) Create a directed version of the Mapper graph. Each undirected edge was assigned a direction from the lower density node to the higher density node. This creates a directed version of the graph that approximates movement towards higher-density regions.
3) Label nodes with states. For each non-maxima node, the directed graph distance d_g was calculated.

   Let B_x: the state of a local maximum V_x and M: set of local maxima

   $$B_x = \{V \mid d_g(V, V_x) < d_g(V, V_y) \ \ \forall\, y \neq x,\ V_y \in M\}$$

   Considerations:
   - Vertices equally closer to a local maxima were labeled as unstable regions, and were not assigned to any particular state.
   - Multiple connected maxima were collapse to the same state.
   - One point might be assigned to more than one state or to a unstable region and one state. Since the states assignment came from the intuition of simulate a gradient Chang et al. interpreted that those points are near to a saddle point "separating" states, and since the "real" coordinates of the saddle point are unknown the data point is assigned to states and/or unstable region with uniform weight.

Temporal Correlation Function

Given that the a system occupied state Bx at time t. Chang et al. defined a temporal correlation as the probability that it will remain (or return) to the state B_x at time $t + \tau$:

$$f_x(t+\tau) = \begin{cases} 1 & \text{if system is associated with } B_x \text{ at time } t+\tau \\ 0 & \text{otherwise.} \end{cases} \tag{4}$$

$$\mathrm{corr}_x(\tau) = \langle f_x(t+\tau) \rangle \tag{5}$$

V. Results

The results below were obtained using the final Mapper parameters reported in Table I. These parameters were selected from the tested parameter combinations because they satisfied the methodological rules described in the Methods section and produced a stable, interpretable graph.

A. The Mapper graph forms a connected compositional landscape

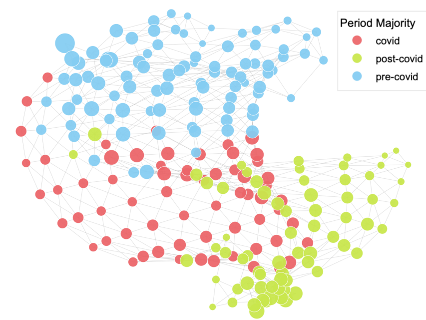

Fig. 3. Mapper graph colored by majority period.

The Mapper graph forms a single connected structure rather than several disconnected components (Fig. 3). I do not interpret this as a failure of the method. In Chang et al., Mapper is used to summarize the shape of a high-dimensional compositional space, not necessarily to force samples into disconnected clusters. In this project, the connected graph suggests that Chicago crime composition changed along a continuous landscape rather than through a clean jump from one regime to another.

At the same time, the period labels are not completely mixed across the graph. Nodes dominated by pre-COVID, COVID, and post-COVID samples tend to occupy different regions of the same connected structure. This suggests gradual compositional reorganization: the city does not appear to move into a single new discrete state, but the temporal periods still leave a visible organization in the landscape.

B. Regions have recognizable profiles

To interpret the regions of the Mapper graph, I colored the same graph using contextual variables that were not used as Mapper inputs (Figs. 4 and 5). For numerical variables, each node was colored by the average value among the samples contained in that node.

The COVID-dominated region has a recognizable profile. It is associated with higher domestic violence rate, higher unemployment rate, and higher poverty rate (Fig. 4). The same area of the landscape also shows higher vacancy rates (vacant housing units, %) (Fig. 5). Arrest rate shows a more mixed pattern: the pre-COVID region contains both high and low values, while the COVID and post-COVID regions tend to show lower values. Median household income shows a related socioeconomic pattern. Pre-COVID and post-COVID areas include a wider mix of income values, while the COVID-dominated region is associated with notably lower income values.

This does not imply that COVID caused these patterns. The result is descriptive and associative. However, it suggests that the COVID associated part of the landscape is not only temporally distinct, but also has a recognizable social and crime profile.

Fig. 4. Mapper graph colored by contextual variables not used as Mapper inputs. COVID-majority regions are associated with higher domestic violence rate, poverty rate, and unemployment rate.

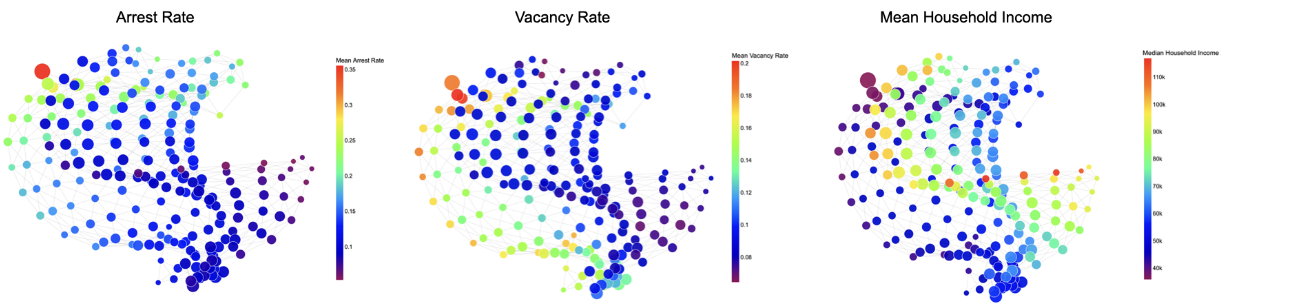

Fig. 5. Mapper graph colored by additional contextual variables. The COVID and post-COVID regions show lower mean arrest rates, while vacancy rate and median household income provide additional socioeconomic context for the observed regions.

C. *Density-based states exist, but they are not isolated clusters*

The local maxima nodes defined candidate states, but these states are not isolated clusters. They are density peaks within a connected graph. In my resultant graph, those peaks exist, but the boundaries are less clear and there is more instability between them (Figure 6). Therefore, I interpret them as density-based neighborhoods rather than hard regimes. In total, 18 states and one unstable region were identified.

Some states are still visually associated with specific temporal regions. For example, State 1 appears mostly in the pre-COVID side of the graph, State 14 is located near the COVID-dominated region, and State 12 appears closer to the post-COVID side. This supports the idea of temporal organization, but not a clean partition of the city into discrete period-specific states.

Figure (7) shows the same point from a community perspective. For this visualization, I retained one state label per community-month in order to draw readable trajectories. The figure contains only 10 randomly selected communities and shows that communities often move across several states within the same period. In other words, even when a single label is assigned for visualization, the periods do not behave like clean blocks where a community stays in one period-specific state.

This reinforces the interpretation of the states as soft density-based neighborhoods. The Mapper graph does not suggest that each community moved cleanly from one state to another. Instead, it suggests that communities moved through a connected compositional landscape where some states became more or less common across periods.

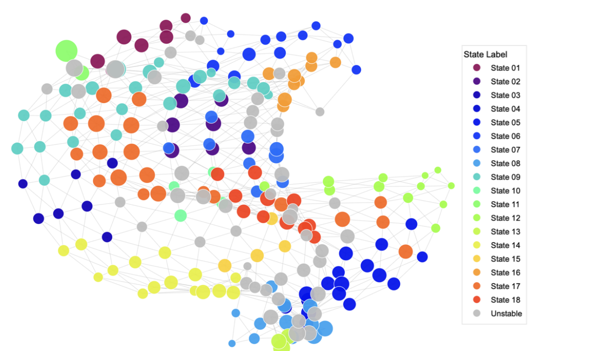

Fig. 6. Mapper graph colored by majority density-based state assignment per node. The states summarize recurring compositional neighborhoods within a connected landscape, while gray nodes indicate unstable or unassigned regions.

D. *Temporal state correlation suggests limited state persistence*

As a final step in the state-based analysis, I computed the temporal correlation function proposed by Chang et al. This measures whether a community that occupies a given state at time t tends to remain in, or return to, that same state after a lag τ. In this project, I interpret the correlation function cautiously because the states are not isolated regimes. In addition, since

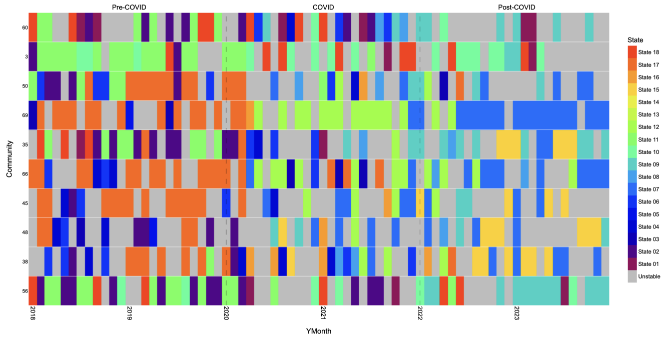

Fig. 7. State trajectories by community and period for 10 randomly selected communities. Each community-month is shown with one retained state label for visualization. The figure shows that communities often move across several states within the same period.

samples can receive partial weight across multiple states, a decay in correlation can mean a real change, but it can also just mean that samples were sitting between two states.

For readability, Figure 8 shows six representative states: two visually associated with the pre-COVID region (01 and 70), two with the COVID region (14 and 18), and two with the post-COVID region (05 and 12). The selected states help summarize the behavior observed in the full interactive exploration.

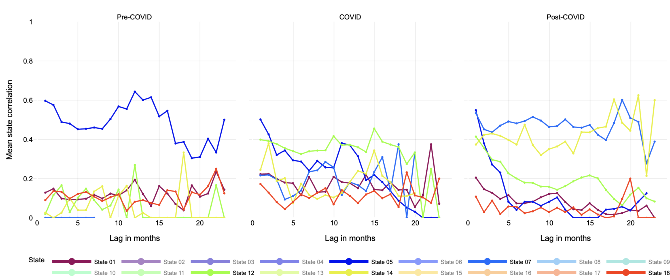

Fig. 8. Temporal state correlation for selected states. The full interactive version was used for exploration, this static figure is included as an illustrative example.

The temporal correlation results reinforce the soft-state interpretation. Some states are visually associated with a period, but they are not exclusive to that period. For example, State 18 is mostly associated with the COVID region, and its correlation decreases with longer lags, suggesting temporary occupation of that neighborhood. State 01 is near the pre-COVID region, but it appears in COVID and post-COVID as well. This is not an error: samples can belong to multiple overlapping nodes and may receive partial association with more than one state. It may also indicate that some profiles were not entirely new, but became more visible or more frequent during one period.

Therefore, this analysis does not show clean state transitions. Instead, it supports the interpretation of the graph as a connected compositional landscape with recurring neighborhoods that become more or less likely across periods.

E. *Community-level shifts are heterogeneous*

The Mapper graph suggested that some regions of the compositional landscape were more associated with specific periods. To explore whether these temporal differences were also visible at the community level, I computed a simple exploratory measure of compositional change for each community area.

For each community and period, I first calculated the mean crime-share vector. I then measured two Jensen-Shannon distances: one between the pre-COVID and COVID mean crime-share vectors, and another between the pre-COVID and post-COVID mean crime-share vectors. These two distances summarize, how much each community changed during the disruption period and how different its post-COVID composition remained from its pre-COVID baseline.

To create an interpretable summary, I used the 0.66 quantile ($q_{0.66}$) of each distance distribution as an exploratory threshold. Communities above this threshold were treated as having relatively large shifts compared with the rest. The ($q_{0.66}$) was used as an exploratory threshold and its purpose is only to flag communities with comparatively stronger compositional changes and to organize the results into qualitative groups.

The shift labels were defined as follows:

- *Temporary Disruption* if its pre-COVID to COVID shift was above the threshold, but its pre-COVID to post-COVID shift was below the threshold. This suggests a disruption during COVID followed by a partial return toward the pre-COVID composition.
- *Persistent Shift* if both distances were above their thresholds, suggesting that the community changed during COVID and remained compositionally different afterward.
- *Delayed Shift* if the pre-COVID to COVID distance was below the threshold, but the pre-COVID to post-COVID distance was above the threshold. This suggests that the largest compositional difference appeared after the COVID period rather than during it.
- *Low Change* when neither distance exceeded its threshold.

This analysis is exploratory and has limitations. It uses period-level mean crime-share vectors, which can smooth within-period variation and hide temporary month-level changes. It also reduces each community's trajectory to two distances, so it does not capture the full path followed by the community through the Mapper landscape. Therefore, these labels should be interpreted as descriptive summaries.

These results suggest that the citywide pattern was not a single uniform transition. Most communities showed low change under this rule, while a smaller subset showed stronger temporary, delayed, or persistent compositional shifts (Fig 9 and Table II). This supports the interpretation of COVID as a heterogeneous disruption rather than a citywide movement into one common post-COVID regime.

The communities labeled as delayed or persistent shifts were then inspected more closely. For readability, Figure 10 shows

TABLE II

EXPLORATORY COMMUNITY-PERIOD SHIFT LABELS.

Shift type	Interpretation	# Communities
Low Change	No large shift in either comparison	37
Temporary Disruption	Large shift during COVID, smaller post-COVID shift	14
Delayed Shift	Larger shift appears after COVID	14
Persistent Shift	Large shift during COVID and post-COVID	12

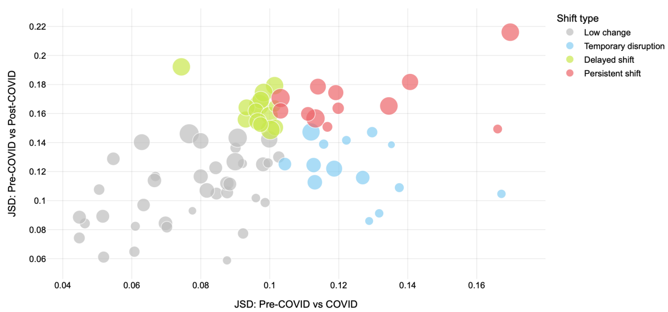

Fig. 9. Exploratory classification of community-level compositional shifts. Each point represents one community area, positioned by its Jensen-Shannon distance from pre-COVID to COVID and from pre-COVID to post-COVID mean crime-share vectors. Colors indicate the shift label assigned using the $q_{0.66}$ thresholds.

the state labels of 14 out of 26 of these communities over time using the state assignment from the full Mapper graph. This figure is a temporal view of how the selected communities moved across the states identified in the full dataset.

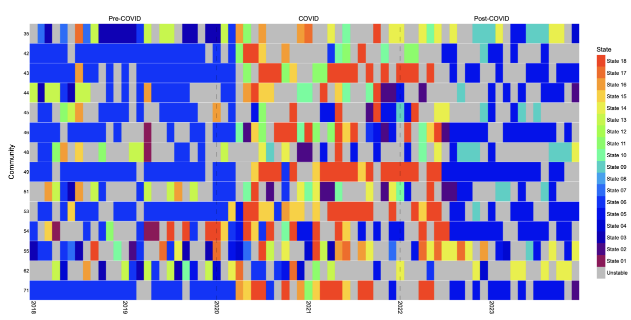

Fig. 10. Temporal state trajectories for the 14 out of 26 delayed or persistent-shift communities. Colors show the state assigned from the full Mapper graph for each community-month, and gray indicates unstable or unassigned nodes. This subset of communities show cleanest paths than communities in Fig 7

F. Exploratory subset analysis of the 26 shifted communities

Motivated by the previous results, I performed an exploratory subanalysis using only the 26 communities labeled as delayed or persistent shifts. These communities were selected because they showed relatively large pre-COVID to post-COVID compositional differences under the JSD-based screening rule. The goal of this subanalysis was not to replace the full Mapper result, but to check whether this subset produced a clearer temporal organization when analyzed on its own.

Applying the same Mapper pipeline (Table I) to this subset produced a smaller landscape with a more visible temporal organization (Fig. 11). Compared with the full graph, the subset graph shows a clearer separation between period-majority regions. This suggests that the global landscape may look continuous partly because many communities changed only moderately, while the selected communities contain a more visible temporal signal.

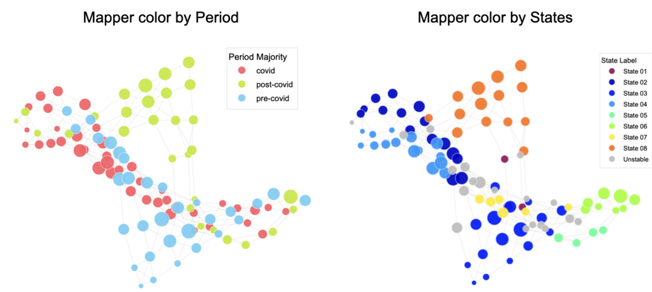

Fig. 11. Exploratory Mapper subset analysis of the 26 communities labeled as delayed or persistent shifts. The left panel shows the subset Mapper colored by majority period, and the right panel shows the same graph colored by state assignment within the subset analysis

Another interesting finding is that the selected communities also showed a spatial and socioeconomic pattern. Most of the 26 delayed or persistent-shift communities are located on the South Side of Chicago (Fig. 12). Compared with the remaining communities, this subset has higher average poverty and unemployment rates, lower median household income, and higher post-COVID domestic violence rates (Table III). Their post-COVID crime composition also shows a sharper increase in motor vehicle theft, while arrest rates remain low.

These patterns do not imply a causal relationship between socioeconomic conditions and compositional shifts. However, they suggest that the communities with stronger post-COVID compositional change were also communities with higher socioeconomic vulnerability. This makes the subset analysis useful as an exploratory follow-up to the full Mapper result.

TABLE III

DESCRIPTIVE COMPARISON OF SHIFTED AND NON-SHIFTED COMMUNITIES

Measure	26 delayed or persistent	Remaining communities
Poverty rate	25%	15%
Unemployment rate	14%	8%
Median household income	$56,317	$81,479
Post-COVID arrest rate	11.81%	10.43%
Post-COVID domestic rate	22.64%	16.84%

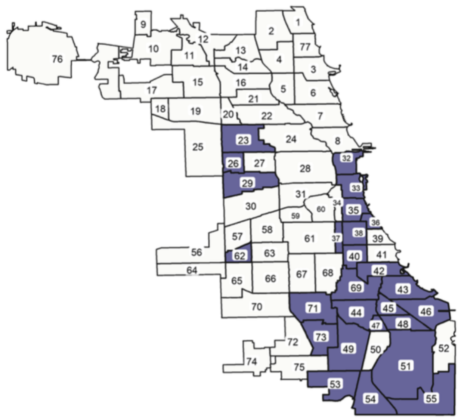

Fig. 12. Geographic location of the 26 communities labeled as delayed or persistent shifts.

VI. DISCUSSION, LIMITATIONS, AND FUTURE WORK

This project asked whether a Mapper-based framework originally developed for microbiome compositional dynamics could be adapted to study crime composition in Chicago before, during, and after COVID-19. The answer is partial. The analysis does not show a clean movement from one pre-COVID state to one COVID state and then to one post-COVID state. Instead, the main result is a connected compositional landscape with visible temporal organization.

This result is still meaningful because the Mapper graph was built only from crime composition. Even with only the relative shares of crime types, pre-COVID, COVID, and post-COVID samples tend to occupy different regions of the same connected structure. This suggests that temporal organization is present in the crime-composition space itself, even if it is not strong enough to form clean separated regimes.

The goal was not to force isolated clusters. The goal was to study whether crime composition could be represented as movement through a landscape. Under that interpretation, the Mapper graph is useful: density-based states summarize recurring compositional neighborhoods; some states are more common in particular periods; and some communities show stronger delayed or persistent shifts than others.

The detected states should therefore be interpreted as soft density neighborhoods rather than hard regimes. This is consistent with the logic of Mapper, where nodes can overlap and a sample can belong to more than one local region. In this project, that overlap is not a technical problem; it is part of the interpretation. It suggests that the city did not switch between clean discrete regimes, but reorganized gradually across a continuous crime-composition space.

There are several reasons why the result may be less clean than the examples in Chang et al. First, urban crime may change more gradually than the biological disruptions studied in the original paper. In microbiome data, an infection can push the system into a very different compositional region. In this dataset, COVID may have redistributed shares among categories without creating a completely new configuration. Second, the crime categories used here are broad. The composition of high-level crime types may not contain enough information to reveal sharper regimes. If the goal is to identify cleaner states, crime composition alone may not be sufficient. Other variables, such as location descriptions, time of day, domestic context, arrest context, or severity, could add more resolution, but incorporating them would require adapting the methodology beyond the purely compositional setup used here. Third, reporting patterns may have shifted during COVID-19. Changes in mobility, police activity, public behavior, and willingness or ability to report incidents may have affected the observed crime composition. Fourth, the community-month scale may smooth short-term shocks. Weekly data or smaller spatial units might show transitions that are hidden after aggregation.

Finally, the selected Mapper setup also favors continuity. The PCoA rank lens, overlapping cover, and single-linkage clustering preserve gradual structure, which is useful for landscape interpretation but makes sharp separation less likely. This is also visible in the PCoA visualization in Appendix B: the projected points already show partial temporal organization, but the regions still overlap substantially.

A limitation of the project is that the analysis is descriptive and does not identify causal effects of COVID-19. The contextual variables, including ACS features, arrest rate, and domestic violence rate, were used only to interpret regions of the Mapper graph. They were not used to build the Mapper graph, and their association with graph regions should not be interpreted causally. Another limitation is that the community-level shift labels depend on an exploratory threshold. The $q_{0.66}$ rule is useful for organizing communities into qualitative groups, but it is not a formal statistical boundary.

Future work could extend the project in several directions. First, the analysis could be repeated with finer temporal or spatial aggregation, such as weekly observations or smaller geographic units, to test whether sharper transitions appear at a more local scale. Second, future versions could incorporate additional crime-context variables that may provide more resolution than crime type shares alone. This would require modifying the current methodology, since the present pipeline is designed for compositional vectors. Third, the robustness of the Mapper graph could be evaluated more formally across distance metrics, filter functions, and hyperparameter choices.

Overall, the research question cannot be answered as a simple yes or no. Chicago community areas did not collectively return to one pre-COVID regime, but they also did not move into one common post-COVID regime. The stronger result is more nuanced: crime composition appears to occupy a continuous landscape with temporal structure, recurring density neighborhoods, and heterogeneous community-level shifts. In that sense, Mapper was useful not because it found clean clusters, but because it gave a structured way to describe gradual and uneven compositional change using composition alone.

Use of AI tools

I used ChatGPT to help debug and optimize the PCoA step. Because the standard Python implementation was too slow for my distance matrix, I implemented a custom function (in the code: `pcoa_fast`) that returns only the first two coordinates needed for the Mapper lens. The optimization was used for computational efficiency and did not change the methodological role of PCoA. Additionally, I used it to create an image (Figure 12). which highlights the Chicago community areas included in the subanalysis.

Appendix A
Jensen-Shannon Distance (JSD)

JSD is a metric defined as the root square of the Jensen Shannon Divergence, which measures the similarity between two probability distributions [7].

$$\mathrm{JSD}(P \,\|\, Q) = \sqrt{\tfrac{1}{2} D_{\mathrm{KL}}(P \,\|\, M) + \tfrac{1}{2} D_{\mathrm{KL}}(Q \,\|\, M)}, \qquad (6)$$

where $M = \frac{1}{2}(P+Q)$ and D_{KL} is the Kullback-Leibler Divergence (KLD).

JSD is a symmetrized and smoothed version of KLD which measures how much an approximating probability distribution Q is different from a true probability distribution P. KLD is not a metric: It does not satisfy symmetry and triangle inequality properties [5].

Kullback-Leibler Divergence [4], [5]:

$$D_{\mathrm{KL}}(P \,\|\, Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}. \qquad (7)$$

Appendix B
PCoA Visualization

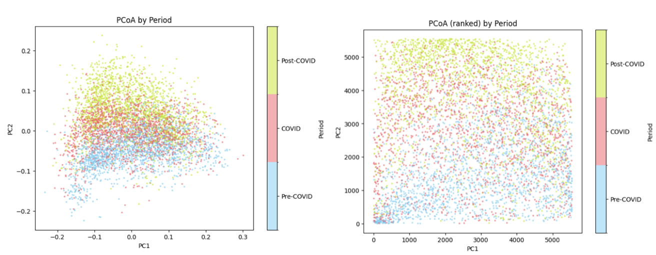

Fig. 13. The raw PCoA view shows partial temporal structure but substantial overlap. After rank transformation, the lens preserves relative ordering while reducing the effect of spacing/outliers. Local clustering was then performed within cover elements using the original distance matrix, not the plotted two-dimensional coordinates.

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